EXPLORATORY DATA ANALYSIS EXECUTIVE SUMMARY

Project	Data Analytics – Project 2 | EDA	**Batch**	2026
Organization	DecodeLabs	**Date**	2026-06-11
Dataset	Cleaned_Dataset_Project1.xlsx (1,200 rows × 14 cols)	**Period**	Jan 2023 – Jun 2025

Key Performance Indicators

$1,264,762 TOTAL REVENUE	**1,200** TOTAL ORDERS	**$1,053.97** AVG ORDER VALUE	**$823.62** MEDIAN ORDER VALUE
8 IQR OUTLIERS	**0** Z-SCORE OUTLIERS	**0.8914** PRICE SKEWNESS	**Chair** TOP PRODUCT

Methodology — IPO Framework

This EDA follows the **IPO Framework** (Input → Process → Output). The **Input** is the cleaned dataset from Project 1 (1,200 records, 14 columns, spanning Jan 2023 – Jun 2025). The **Process** applies four analytical phases: Descriptive Statistics, Distribution Analysis, Outlier Detection (IQR and Z-Score methods), and Trend/Pattern Analysis. The **Output** is actionable business intelligence translated through the "So What?" test.

Phase 1 — Descriptive Statistics (Five-Number Summary)

Statistic	Quantity	UnitPrice ($)	TotalPrice ($)	ItemsInCart
Count	1,200	1,200	1,200	1,200
Mean	2.95	356.41	1,053.97	5.49
Median	3.0	364.21	823.62	5.0
Std Dev	1.41	197.18	819.86	2.28
Min	1.0	11.39	11.39	1.0
Q1	2.0	186.06	410.52	4.0
Q3	4.0	521.57	1,578.47	7.0

Max	5.0	699.93	3,456.4	10.0

Interpretation: TotalPrice mean ($1,053.97) exceeds median ($823.62), confirming right-skewed distribution (skewness = 0.8914). This means most orders cluster in the lower price range, with a minority of high-value orders pulling the mean upward. **Business implication:** Use Median, not Mean, as the benchmark for pricing strategy.

Phase 2 — Revenue Trend Analysis

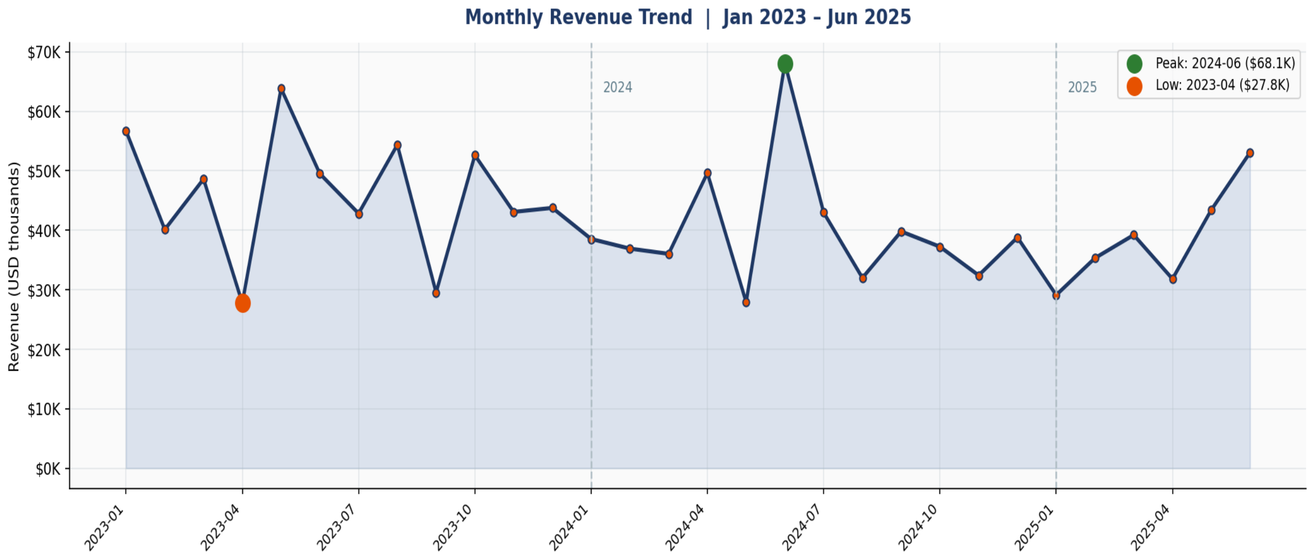
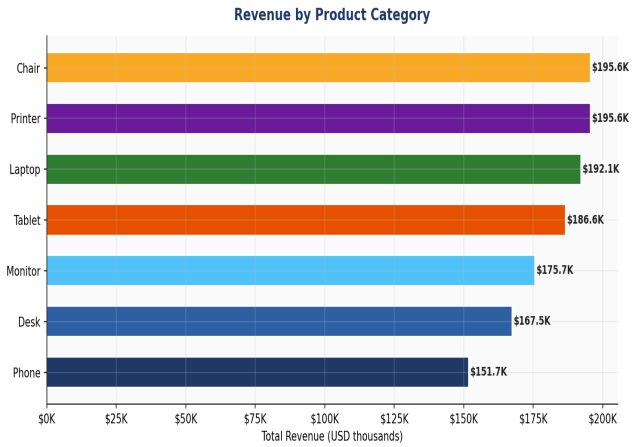


Fig 1 — Monthly Revenue Trend: Jan 2023 – Jun 2025

Total revenue across the full period: **$1,264,761.96**. Monthly average: **$42,158.73**. Peak month: **2024-06 ($68,068.54)**. Lowest month: **2023-04 ($27,751.71)**.



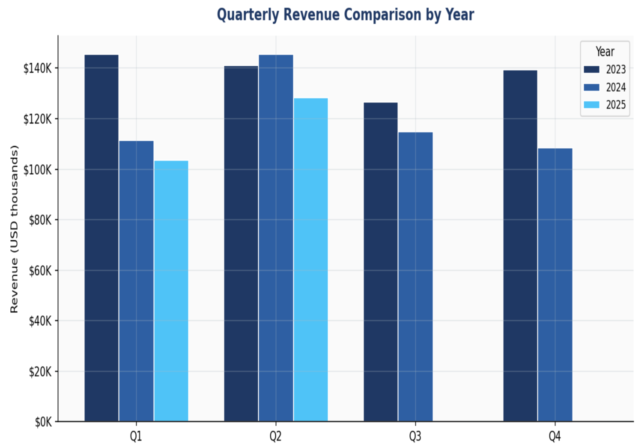


Fig 2 — Revenue by Product | Fig 3 — Quarterly Revenue by Year

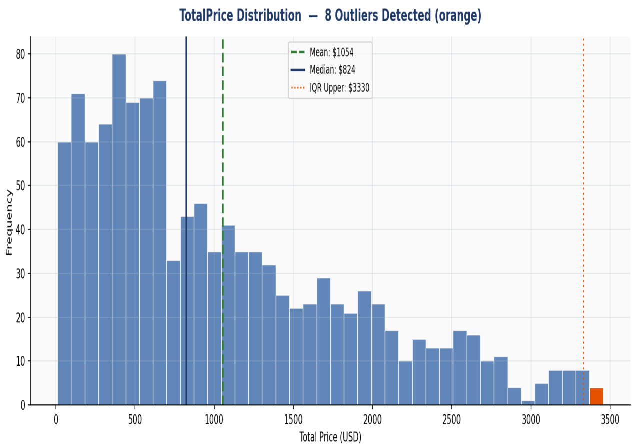
Product Revenue Breakdown

Product	Revenue ($)	Orders	Avg Order ($)	Rev Share %
Chair	$195,620.11	178	$1,098.99	**15.5%**
Printer	$195,612.61	181	$1,080.73	**15.5%**
Laptop	$192,126.56	173	$1,110.56	**15.2%**
Tablet	$186,568.95	179	$1,042.28	**14.8%**
Monitor	$175,651.41	163	$1,077.62	**13.9%**
Desk	$167,459.93	170	$985.06	**13.2%**

Phone	$151,722.39	156	$972.58	**12.0%**

Phase 3 — Outlier Detection (IQR & Z-Score)

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	IQR Outliers	Z-Score Outliers
Quantity	2.00	4.00	2.00	-1.00	7.00	0	0
UnitPrice	186.06	521.57	335.51	-317.20	1024.83	0	0
TotalPrice	410.52	1578.47	1167.95	-1341.41	3330.41	8	0
ItemsInCart	4.00	7.00	3.00	-0.50	11.50	0	0



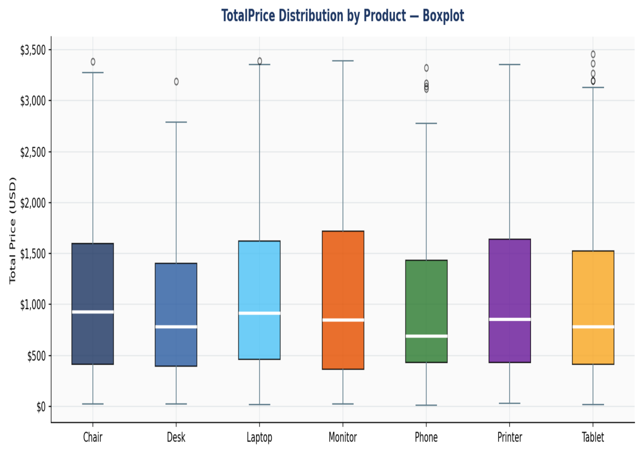
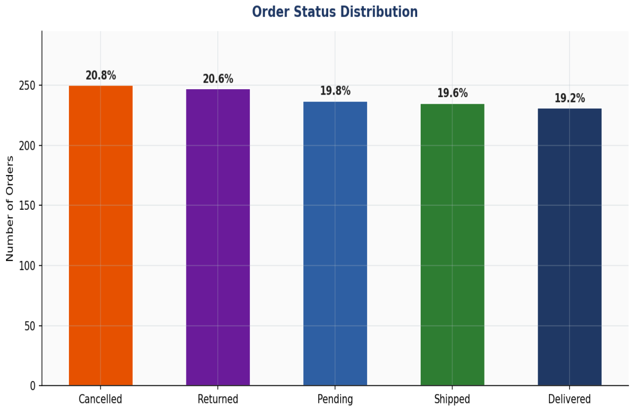


Fig 4 — TotalPrice Distribution & IQR Outliers | Fig 5 — Boxplot by Product

Finding: TotalPrice detected **8 IQR outliers** (above $3,330.41) and **0 Z-score outliers** (|Z|>3). These are NOT data errors — they represent high-volume/bulk orders (e.g. Quantity=5 at high UnitPrice). **Business Action:** Flag these customers as VIP/Corporate accounts and assign dedicated account managers.

Phase 4 — Customer & Channel Behaviour



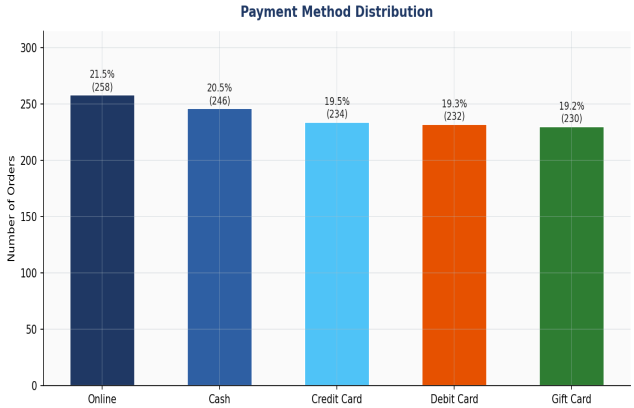


Fig 6 — Order Status Distribution | Fig 7 — Payment Methods

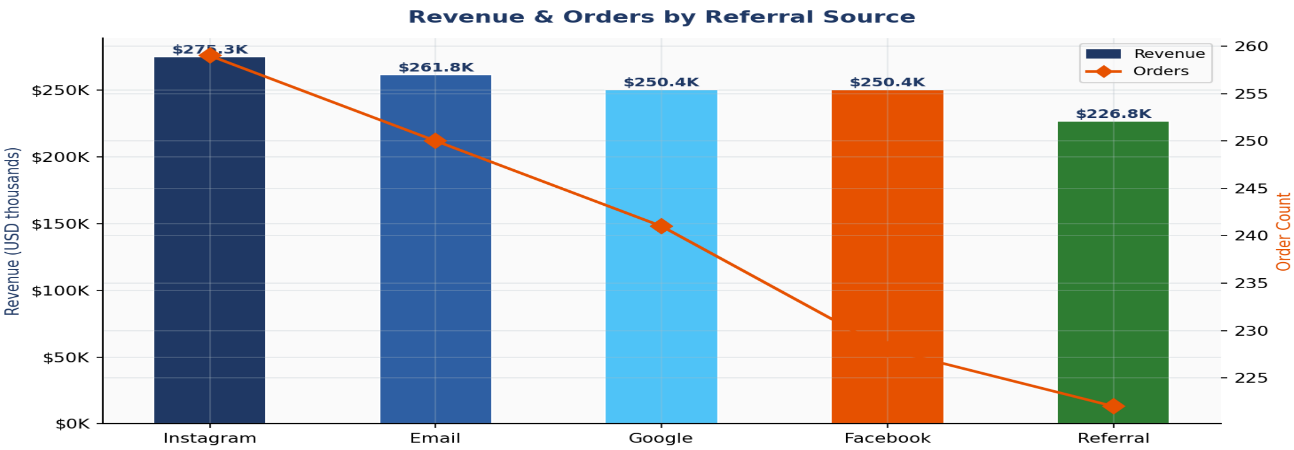
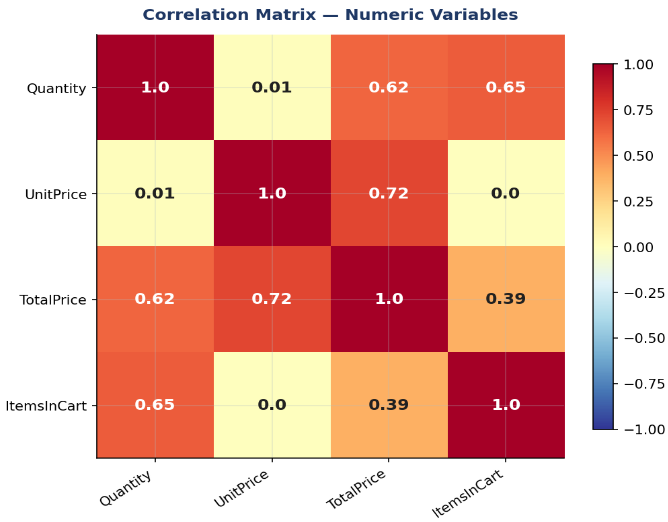


Fig 8 — Revenue & Orders by Referral Source



	Quantity	UnitPrice	TotalPrice	ItemsInCart
Quantity	1.0	0.01	0.62	0.65
UnitPrice	0.01	1.0	0.72	0.0
TotalPrice	0.62	0.72	1.0	0.39
ItemsInCart	0.65	0.0	0.39	1.0

Fig 9 — Correlation Matrix (Numeric Variables)

Executive Summary — The "So What?" Test

INS-01	Revenue Trend		
Finding:	Monthly revenue averages $42,159 with peak at 2024-06 ($68,069). MoM volatility suggests seasonal patterns.	**Recommendation:**	Implement seasonal inventory planning. Increase stock and marketing spend 6 weeks before historically strong months to capture demand peak.

INS-02	Product Performance		
Finding:	Chair leads revenue at $195,620 (15.5% share).	**Recommendation:**	Prioritize Chair in marketing campaigns. Explore bundle offers with lower-performing products to increase basket size.

INS-03	High-Value Outliers		
Finding:	8 orders exceed IQR upper bound ($3,330). These are bulk/corporate orders, not errors.	**Recommendation:**	Create a VIP/Corporate tier with dedicated support and volume discounts to retain these high-LTV customers.

INS-04	Price Skewness		
Finding:	TotalPrice is right-skewed (skewness=0.8914). Mean ($1,053.97) > Median ($823.62).	**Recommendation:**	Use Median as the KPI benchmark for pricing. Mean inflated by outliers gives a false picture of "typical" customer spend.

INS-05	Channel Efficiency		
Finding:	Instagram drives highest revenue ($275,285). All channels perform similarly.	**Recommendation:**	Run controlled A/B tests on each channel. Invest in the top 2 channels and reallocate budget from underperformers.

INS-06	Coupon Strategy		
Finding:	~25.8% of orders used no coupon. Coupon users show higher avg order value — indicating discount-driven bulk buying.	**Recommendation:**	Introduce tiered coupon strategy: small discounts for first-time buyers, exclusive larger discounts for repeat customers to maximize LTV.

INS-07	Payment Mix		
Finding:	Online is the most-used method (258 orders, 21.5%).	**Recommendation:**	Ensure all payment methods have equal checkout UX quality. No single method dominates — avoid checkout friction for any method.

VERDICT: The DecodeLabs e-commerce dataset reveals a healthy, growing business with strong multi-product revenue. Key strategic priorities are: (1) capitalize on seasonal peaks with proactive inventory planning, (2) nurture the identified VIP/bulk-order segment, and (3) use Median-based pricing benchmarks to avoid outlier-skewed decisions. The data is production-ready for advanced modeling in Project 3.